Examples: Some Important Results Based on gamma & beta distributions

I) If $X \sim \Gamma_\lambda$ (one parameter)

$$f(x) = \frac{1}{\Gamma\lambda} e^{-x} x^{\lambda-1} \; ; \; \lambda > 0, \; 0 < x < \infty$$

$$M_X(t) = (1-t)^{-\lambda}, \; |t| < 1$$

Mean $= \lambda$, Variance $= \lambda$ (Mean = Variance)

II) $X \sim \Gamma_{a,\lambda}$ (two parameter)

$$f(x) = \frac{a^\lambda}{\Gamma\lambda} e^{-ax} x^{\lambda-1}, \quad a > 0, \lambda > 0, \; 0 < x < \infty$$

Remark: $X \sim \Gamma_\lambda = \Gamma(1,\lambda)$

$$M_X(t) = \left(1 - \frac{t}{a}\right)^{-\lambda} ; \quad t < a$$

Mean $= \frac{\lambda}{a}$, variance $= \frac{\lambda}{a^2} = \frac{\text{Mean}}{a}$

Hence, Variance > Mean if $a < 1$

Variance = Mean if $a = 1$

Variance < Mean if $a > 1$

III) $X \sim \beta_1(\mu, \nu)$

$$f(x) = \frac{1}{\beta(\mu,\nu)} x^{\mu-1} (1-x)^{\nu-1} \; ; \; (\mu,\nu) > 0, \quad 0 < x < 1$$

Mean $= \frac{\mu}{\mu+\nu}$, Variance $= \frac{\mu\nu}{(\mu+\nu)^2(\mu+\nu+1)}$

IV) $X \sim \beta_2(\mu,\nu)$

$$f(x) = \frac{1}{\beta(\mu,\nu)} \cdot \frac{x^{\mu-1}}{(1+x)^{\mu+\nu}} \; ; \; (\mu,\nu) > 0, \; 0 < x < \infty$$

Remark: If $X \sim \beta_2(\mu,\nu)$, $y = \frac{1}{1+x}$, $Y \sim \beta_1(\mu,\nu)$

Mean $= \frac{\mu}{\nu-1}$, $\nu > 1$

Variance $= \frac{\mu(\mu+\nu-1)}{(\nu-1)^2(\nu-2)}$

Remarks :

i) if $X_i \sim \Gamma\lambda_i$ $(i=1,2,\dots n)$ are independent Γ variates
$\sum_{1}^{n} X_i \sim \Gamma \sum_{1}^{n}\lambda_i$

ii) if $X_i \sim \Gamma(a,\lambda_i)$ $(i=1,2,\dots n)$ are independent gamma variates.
$\sum_{1}^{n} X_i \sim \Gamma(a, \sum_{1}^{n}\lambda_i)$

Results : $X \sim \Gamma(\lambda, a)$ and $Y \sim \Gamma(\lambda, b)$ be independent random variables.

Show that $E\left(\frac{X}{X+Y}\right) = \frac{E(X)}{E(X+Y)}$

Hint : Since X and Y are independent r.v's, then $U = \frac{X}{X+Y}$ and $V=(X+Y)$ are independent.

Hence, $E(UV) = E(U)E(V) \Rightarrow E(X) = E\left(\frac{X}{X+Y}\right)\cdot E(X+Y)$

$\Rightarrow \frac{E(X)}{E(X+Y)} = E\left(\frac{X}{X+Y}\right)$

Results : if $X \sim \Gamma(\lambda, \mu)$ and $Y \sim \Gamma(\lambda, \nu)$ are independent r.v's then show that

$\frac{X}{Y} \sim \beta_2(\mu, \nu)$ and $\frac{X}{X+Y} \sim \beta_1(\mu, \nu)$

Ex

If $X \sim \Gamma(\lambda, \mu)$, Mean $= 6$, variance $= 3$
$Y \sim \Gamma(\lambda, \nu)$, Mean $= 8$, variance $= 4$

Find $E\left(\frac{X}{X+Y}\right)$

We know if $X \sim \Gamma(a, \lambda)$, Mean $= \frac{\lambda}{a}$, variance $= \frac{\lambda}{a^2}$

Hence for $X \sim \Gamma(\lambda, \mu)$

given $\frac{\mu}{\lambda} = 6$, $\frac{\mu}{\lambda^2} = 3$ gives $6\lambda = 3\lambda^2 \Rightarrow \lambda = 2$
and $\mu = 12$

Similarly for $Y \sim \Gamma(\lambda, \nu)$

$$\text{Mean} = \frac{\nu}{\lambda}, \quad \text{Variance} = \frac{\nu}{\lambda^2}$$

Hence $\frac{\nu}{\lambda} = 8, \quad \frac{\nu}{\lambda^2} = 4 \implies 8\lambda = 4\lambda^2$

$\lambda = 2, \nu = 16$

Hence $X \sim \Gamma(2, 12)$

$Y \sim \Gamma(2, 16) \implies \frac{X}{X+Y} \sim \beta_1(12, 16)$

$$E\left(\frac{X}{X+Y}\right) = \text{Mean of } \left(\frac{X}{X+Y}\right) = \frac{12}{12+16} = \frac{12}{28} = \frac{3}{7}.$$

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Regression Curves

The conditional mean $E(Y/X = x)$ for continuous distn is called regression function of Y on X and the graph of this function of x is known as the regression curve on Y on X or regression curve for the mean of Y.

Similarly, the regression function of X on Y is $E(X/Y = y)$ and the graph of this function of y is called the regression curve (of the mean) of X on Y.

In case a regression curve is a straight line, the corresponding regression is said to be linear. If one of the regression is linear, it does not however follow that other is also linear.

Theorem: Let (X, Y) be a two-dimensional random variable with $E(X) = \bar{X}$, $E(Y) = \bar{Y}$, $V(X) = \sigma_x^2$, $V(Y) = \sigma_y^2$ and r_{xy} be the correlation coefficient between X and Y, if the regression of Y on X is linear, then

$$E(Y/X) = \bar{Y} + r\frac{\sigma_y}{\sigma_x}(X - \bar{X})$$

Similarly, if the regression of X on Y is linear, then

$$E(X/Y) = \bar{X} + r\frac{\sigma_x}{\sigma_y}(Y - \bar{Y}).$$

Ex. Given $f(x,y) = x e^{-x(y+1)}$; $x \geq 0, y \geq 0$

find the regression curve of Y on X.

Soln: Marginal p.d.f. of X is given by

$$f_1(x) = \int_0^\infty f(x,y)\,dy = \int_0^\infty x e^{-x(y+1)}\,dy$$

$$= e^{-x}, \quad x \geq 0$$

Conditional p.d.f. of Y ~~for given~~ on X is given by

$$f(y/x) = \frac{f(x,y)}{f_1(x)} = \frac{x e^{-x(y+1)}}{e^{-x}} = x e^{-xy}$$

The regression curve of Y for given X is given by

$$y = E(Y/X=x) = \int_0^\infty y \cdot f(y/x)\,dy = \int_0^\infty y x e^{-xy}\,dy$$

$$= \frac{1}{x}$$

i.e $y = \frac{1}{x} \Rightarrow xy = 1$, which is ~~the~~ not linear. Hence, the regression of Y on X is not linear.

Ex. Obtain regression equation of Y on X for the following distribution.

$$f(x,y) = \frac{y}{(1+x)^4} \exp\left(-\frac{y}{1+x}\right); \quad (x,y) \geq 0$$

Marginal p.d.f. of X is given by

$$f_1(x) = \int_0^\infty f(x,y)\,dy = \frac{1}{(1+x)^4} \int_0^\infty y e^{-\frac{y}{(1+x)}}\,dy$$

$$= \frac{1}{(1+x)^4} \sqrt{2}\,(1+x)^2$$

$$= \frac{1}{(1+x)^2}; \quad x \geq 0$$

[using Γ integral $\int_0^\infty x^{n-1} e^{-ax}\,dx = \frac{\Gamma n}{a^n}$]

The conditional p.d.f. of Y (for given X) is given by:

$$f(y/x) = \frac{f(x,y)}{f_1(x)} = \frac{y}{(1+x)^2}\exp\left(-\frac{y}{1+x}\right);\ y \geq 0$$

Regression eqn of Y on X is given by

$$y = E(Y/X) = \int_0^\infty y\, f(y/x)\, dy = \int_0^\infty \frac{1}{(1+x)^2}\, y^2 e^{-y/(1+x)}\, dx$$

$$= \frac{1}{(1+x)^2}\sqrt{3}\,(1+x)^3 \quad \text{(using gamma integral)}$$

$$= 2(1+x)$$

$\therefore$ $y = 2(1+x)$, Hence the regression of Y on X is linear.

Ex Variables X and Y have the joint p.d.f. given by:

$$f(x,y) = \frac{1}{3}(x+y);\ 0 \leq x \leq 1,\ 0 \leq y \leq 2.$$

Find (i) r_{xy} (ii) The two lines of regression

Soln The marginal p.d.f.s of X and Y are

$$f_1(x) = \int_0^2 f(x,y)\,dy = \int_0^2 \frac{1}{3}(x+y)\,dy = \frac{2}{3}(1+x);\ 0 \leq x \leq 1$$

$$f_2(y) = \int_0^1 f(x,y)\,dx = \int_0^1 \frac{1}{3}(x+y)\,dx = \frac{1}{3}\left(\frac{1}{2}+y\right);\ 0 \leq y \leq 2$$

$$E(X) = \int_0^1 x f_1(x)\,dx = \frac{5}{9} \qquad E(X^2) = \int_0^1 x^2 f_1(x)\,dx = \frac{7}{18}$$

$$\Rightarrow V(X) = E(X^2) - (E(X))^2 = \frac{13}{162} \quad \text{Similarly } V(Y) = \frac{23}{81}$$

$$E(XY) = \int_0^1\int_0^2 xy\, f(x,y)\,dx\,dy = \frac{2}{3}$$

$$Cov(X,Y) = E(X,Y) - E(X)E(Y) = -\frac{1}{81}$$

$$r_{xy} = \frac{Cov(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}} = -\sqrt{\left(\frac{2}{299}\right)}$$

$\bar{X} = E(X)$
$\bar{Y} = E(Y)$

The two lines of regression are $Y = \bar{Y} + \frac{Cov(X,Y)}{\sigma_x^2}(X - \bar{X})$

$$= \frac{11}{9} - \frac{2}{13}\left(X - \frac{5}{9}\right)$$

Similarly and $X = \frac{5}{9} - \frac{1}{23}\left(Y - \frac{11}{9}\right)$